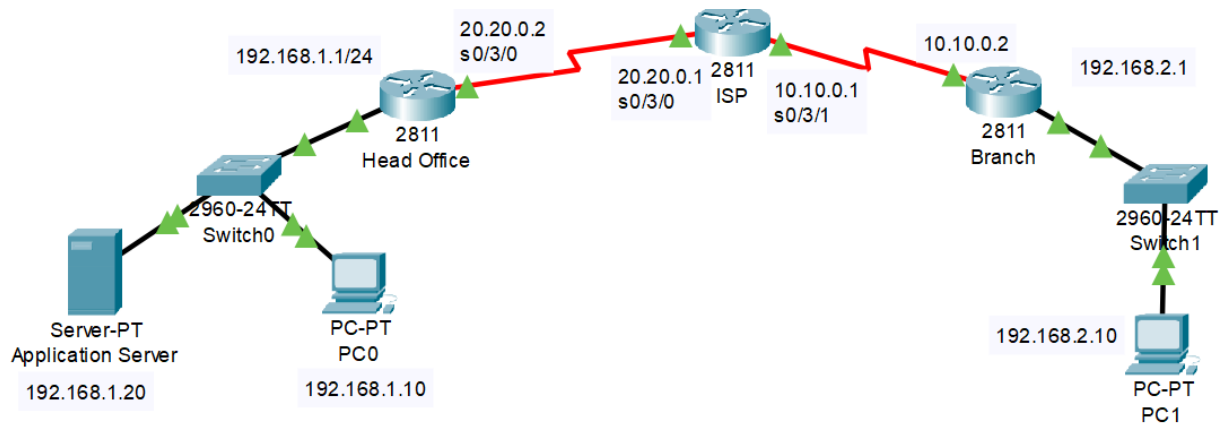


Site to Site VPN Configuration

1. Draw necessary topology
2. Configure IP address to the Router interfaces and the PCs
3. Configure Routing protocol/ Static route and test communication
4. Configure IPsec VPN
 - Enable security technology package
 - Configure extended ACL permitting the target on each router
 - Configure the IKE Phase1 ISAKMP policy on each router
 - Configure the IKE Phase 2 IPsec policy on each router
 - Configure the crypto map on the outgoing interface.
 -



Enable security technology package

```
Router(config)#license boot module c2900 technology-package securityk9
```

Test

Show version

```
Security >>> security k9
```

Router HQ

```
Router(config)#host
```

```
Router(config)#hostname HQ
```

```
HQ(config)#inter
```

```
HQ(config)#interface f0/0
```

```
HQ(config-if)#ip add
HQ(config-if)#ip address 192.168.1.1 255.255.255.0
HQ(config-if)#no shut
HQ(config-if)#no shutdown
HQ(config-if)#exit
```

```
HQ(config)#inter s0/3/0
HQ(config-if)#ip add
HQ(config-if)#ip address 20.20.0.2 255.255.255.0
HQ(config-if)#no shut
HQ(config-if)#no shutdown
```

```
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
HQ(config-if)#
```

Remote Site/ Branch

Router

```
Router(config)#hostname Branch
```

```
Branch(config)#int f0/0
Branch(config-if)#ip add
Branch(config-if)#ip address 192.168.2.1 255.255.255.0
Branch(config-if)#no shut
Branch(config-if)#no shutdown
```

```
Branch(config-if)#int s0/3/0
Branch(config-if)#ip add
Branch(config-if)#ip address 10.10.0.2 255.255.255.0
Branch(config-if)#no shut
Branch(config-if)#no shutdown
```

```
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
Branch(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up
```

```
Branch# show ip interface brief
```

ISP

```
Router(config)#hostname ISP
```

```
ISP(config)#
```

```
ISP(config)#
```

```
ISP(config)#inter
```

```
ISP(config)#interface s0/3/0
```

```
ISP(config-if)#ip add
```

```
ISP(config-if)#ip address 20.20.0.1 255.255.255.0
```

```
ISP(config-if)#int s0/3/1
```

```
ISP(config-if)#ip add
```

```
ISP(config-if)#ip address 10.10.0.1 255.255.255.0
```

```
ISP(config-if)#no shut
```

```
ISP(config-if)#no shutdown
```

```
ISP(config-if)#
```

```
%LINK-5-CHANGED: Interface Serial0/3/1, changed state to up
```

```
ISP(config-if)#int s0/3/0
```

```
ISP(config-if)#ip address 20.20.0.1 255.255.255.0
```

```
ISP(config-if)#no shut
```

```
ISP(config-if)#no shutdown
```

```
ISP(config-if)#
```

```
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up
```

Pc0

```
Ping 192.168.1.1>>> Reply ok
```

```
Ping 192.168.2.1 >>> not ok
```

Static Route

```
ISP(config)#ip route 192.168.1.0 255.255.255.0 20.20.0.2
```

```
ISP(config)#ip route 192.168.2.0 255.255.255.0 10.10.0.2
```

```
ISP(config)#
```

HQ

```
HQ(config)#ip route
```

```
HQ(config)#ip route 0.0.0.0 0.0.0.0 20.20.0.1
```

```
HQ(config)#
```

Branch

```
Branch(config)#ip route
Branch(config)#ip route 0.0.0.0 0.0.0.0 10.10.0.1
Branch(config)#
```

IPSec tunnel on HQ

```
HQ(config)#crypto isakmp enable
HQ(config)#crypto isakmp policy ?
<1-10000> Priority of protection suite
HQ(config)#crypto isakmp policy 20 ?
<cr>
HQ(config)#crypto isakmp policy 20
HQ(config-isakmp)#authen
HQ(config-isakmp)#authentication pre
HQ(config-isakmp)#authentication pre-share
```

```
HQ(config-isakmp)#encryption 3des
```

```
HQ(config-isakmp)#hash md5
```

```
HQ(config-isakmp)#group 1
```

```
HQ(config-isakmp)#lifetime 3600
HQ(config-isakmp)#exit
```

```
HQ(config)#crypto isakmp key
HQ(config)#crypto isakmp key ?
WORD The UNENCRYPTED (cleartext) user password
```

```
HQ(config)#crypto isakmp key bcas123 add
HQ(config)#crypto isakmp key bcas123 address 10.10.0.2
```

```
HQ(config)#crypto ipsec transform-set ?
WORD Transform set tag
HQ(config)#crypto ipsec transform-set myset ?
ah-md5-hmac AH-HMAC-MD5 transform
ah-sha-hmac AH-HMAC-SHA transform
esp-3des ESP transform using 3DES(EDE) cipher (168 bits)
esp-aes ESP transform using AES cipher
esp-des ESP transform using DES cipher (56 bits)
esp-md5-hmac ESP transform using HMAC-MD5 auth
```

esp-sha-hmac ESP transform using HMAC-SHA auth

```
HQ(config)#crypto ipsec transform-set myset esp-3des esp-md5-hmac
HQ(config)#
```

Access-List

```
HQ(config)#access-list 100 permit ip 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255
```

```
HQ(config)#crypto map ?
WORD Crypto map tag
ipv6 IPv6 crypto map
HQ(config)#crypto map mymap
HQ(config)#crypto map mymap ?
<1-65535> Sequence to insert into crypto map entry
client Specify client configuration settings
isakmp Specify isakmp configuration settings
HQ(config)#crypto map mymap 20 ipse
HQ(config)#crypto map mymap 20 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
HQ(config-crypto-map)#set pe
HQ(config-crypto-map)#set peer 10.10.0.2
```

```
HQ(config-crypto-map)#set transform-set myset
HQ(config-crypto-map)#match
HQ(config-crypto-map)#match add
HQ(config-crypto-map)#match address ?
<100-199> IP access-list number
WORD Access-list name
HQ(config-crypto-map)#match address 100
HQ(config-crypto-map)#exit
HQ(config)#
```

```
HQ(config-crypto-map)#match address 100
HQ(config-crypto-map)#exit
HQ(config)#inter
HQ(config)#interface s0/3/0
HQ(config-if)#cryp
HQ(config-if)#crypto map
HQ(config-if)#crypto map mymap
*Jan 3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
HQ(config-if)#
```

Configure ipsec Tunnel on Branch

Branch#config t

Enter configuration commands, one per line. End with CNTL/Z.

Branch(config)#crypto isakmp enable

Branch(config)#crypto isakmp policy 20

Branch(config-isakmp)#authentication pre-share

Branch(config-isakmp)#encryption 3des

Branch(config-isakmp)#hash md5

Branch(config-isakmp)#group 1

Branch(config-isakmp)#lifetime 3600

Branch(config-isakmp)#exit

Branch(config)#crypto isakmp key bcas123 address 20.20.0.2

Branch(config)#crypto ipsec transform-set myset esp-3des esp-md5-hmac

Branch(config)#access-list 100 permit ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255

Branch(config)#crypto map mymap 20 ipsec-isakmp

% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.

Branch(config-crypto-map)#set peer 20.20.0.2

Branch(config-crypto-map)#set transform-set myset

Branch(config-crypto-map)#match address 100

Branch(config-crypto-map)#exit

Branch(config)#

Branch(config)#interface s0/3/0

Branch(config-if)#crypto map mymap

*Jan 3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON

Branch(config-if)#

HQ#show cry

HQ#show crypto isa

HQ#show crypto isakmp s

HQ#show crypto isakmp sa

IPv4 Crypto ISAKMP SA

dst src state conn-id slot status

10.10.0.2 20.20.0.2 QM_IDLE 1078 0 ACTIVE (deleted)

IPv6 Crypto ISAKMP SA

HQ#show crypto isakmp sa

IPv4 Crypto ISAKMP SA

dst src state conn-id slot status

10.10.0.2 20.20.0.2 QM_IDLE 1078 0 ACTIVE (deleted)

IPv6 Crypto ISAKMP SA

```
HQ#show crypto isakmp poli
HQ#show crypto isakmp policy
```

```
Global IKE policy
Protection suite of priority 20
encryption algorithm: Three key triple DES
hash algorithm: Message Digest 5
authentication method: Pre-Shared Key
Diffie-Hellman group: #1 (768 bit)
lifetime: 3600 seconds, no volume limit
HQ#show crypto ma
```

```
HQ#show crypto map
Crypto Map mymap 20 ipsec-isakmp
Peer = 10.10.0.2
Extended IP access list 100
access-list 100 permit ip 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255
Current peer: 10.10.0.2
Security association lifetime: 4608000 kilobytes/3600 seconds
PFS (Y/N): N
Transform sets={
myset,
}
Interfaces using crypto map mymap:
Serial0/3/0
```